

Response to ASTRA Proposal Queries

From: BKS Solutions **To:** ASTRA — Procurement / IT / Commercial review **Subject:** Response to eight queries raised on the Procurement Comparison Portal proposal **Date:** 18 August 2026 **Status:** For discussion. Commercial figures marked *indicative* are subject to the commercial schedule.

Purpose and posture

Eight queries were raised. Three of them identify things our proposal did not state clearly enough, and we accept those without argument. Four require information we have but did not put in front of you. One — the proposal to run a measured POC on real ASTRA data before committing — is the right next step and we agree with it outright.

We have written this document to be checkable rather than reassuring. Where a capability is built, we say what it is and how you can verify it during the POC. Where a capability is **not** built, we say so in the same sentence rather than in a footnote. Section 9 lists every such gap in one place, because a proposal that only lists strengths is not one you can plan against.

Summary of responses

#	Query	Our position	Where
1	Extraction/comparison portal, not full procurement	Partly accepted. The deck under-represented the platform — the eight-stage RFQ lifecycle is built. But it is not a source-to-pay suite and we will not claim it is.	\$1
2	SaaS vs on-premise confusion	Accepted. The proposal mixed two delivery models.	\$2
3	AI architecture, residency, cost, model dependency, security undefined	Accepted.	\$3, Appendix B
4	Customisation cost needs a deliverables matrix	Accepted.	\$4, Appendix A
5	10 named users impractical across four functions	Accepted.	\$5
6	Engineering AI scope too broad	Accepted — withdrawn from Phase 1.	\$6
7	Ownership, source code, data, prompts, exit	Answerable in full.	\$7
8	Run a measured POC first	Agreed.	\$8, Appendix C

1. "Mainly an AI document extraction and comparison portal rather than a complete procurement solution"

What is fair in this

The pitch deck we sent describes seven steps, and all seven of them are the comparison pipeline: intake, recognition, lineage, reading, filing, measurement, decision. Read on its own, the deck supports exactly the conclusion you reached. That is a failure of our document, not a misreading on your side.

What the deck omitted

The platform is two subsystems, not one. Alongside the comparison pipeline there is a complete RFQ lifecycle, built and under test, exposed through 47 HTTP endpoints:

Capability	State	Notes
Eight-stage RFQ lifecycle	Built	Shortlisting → Issued → Clarifications → Bids Received → Evaluation → Negotiation → Awarded → PO Issued. Fixed process codes RFQ-02 ... RFQ-07, deny-by-default stage transitions, and documented backward recoveries (retender, renegotiate).
Project → item → RFQ hierarchy	Built	An RFQ covers named line items within a project; deletion is refused while anything live references the target.
Bidder registry	Built	Organisation-wide, SQLite-backed, ~1,300 vendors folded from a real ADNOC Approved Vendor List export, plus an Astra-approved subset. Queried by discipline, product group and approval.
Vendor eligibility / prequalification	Built	Suitability computed from prequal status, expiry and scope fit. Expiry is derived at read time against a date the caller passes in — there is no overnight sweep that can leave the register stale.
Shortlisting and client approval	Built	Per-vendor invitation as an attributed act; inviting a blocked bidder requires a recorded override reason. Client approval is derived live from the AVL, so correcting one registry row corrects every shortlist that references it.
Draft shortlists per item	Built	A buyer assembles candidates before any RFQ exists; adoption into invitations happens when the RFQ is raised, with every skipped vendor reported by name and reason.
Clarification rounds	Built	Numbered queries, computed Open/Answered/Withdrawn state, restricted answers with a mandatory recorded reason, and grounded answer drafting from the enquiry package.
Addenda	Built	The single sanctioned route through the frozen-package rule; supersedes at a new revision, with concurrency guards.
Enquiry and bid document management	Built	Content-addressed storage. One collection holds both the enquiry the contractor issued and the documents bidders returned.
Technical bid evaluation (TBE) / VDRL	Built (server)	Routes served and tested; some browser controls were deliberately removed — see §9.
Discipline vocabulary	Built	Controlled vocabulary joined to the client's own product-group taxonomy, so scope matching resolves against the real register rather than free text.
Comparison pipeline	Built	The seven steps in the deck.
Cost estimation	Built (CLI only)	A separate costing-sheet ingestion CLI. Not reachable from the web application today.

What it is genuinely not

Stated directly, so that nobody plans against something that does not exist:

- **It is not a source-to-pay suite.** There is no PO issuance to an ERP, no goods receipt, no invoice matching, no payment, no contract lifecycle management, no supplier performance scorecarding, no e-auction, and no catalogue or punch-out.
- **Nothing sends mail.** Recipient addresses are recorded; there is no mail transport. An RFQ, an answer or an addendum is produced by the platform and issued by a person today. This is a named customisation item (Appendix A, C-3).
- **No ERP integration is built.** SAP/Oracle/other integration is a defined customisation deliverable, not a shipped feature.

Proposed correction

We will re-issue the proposal with the scope stated as **"RFQ management and bid evaluation"** rather than as a comparison portal, with the boundary above printed in the document rather than discovered in review. If ASTRA's requirement is genuinely source-to-pay, we would rather establish that now than in month four.

2. SaaS subscription vs on-premise deployment — where AI processing and data reside

Accepted without qualification

The proposal used SaaS subscription language ("seats", "credits", "plans") alongside security language about deploying "inside your perimeter". Those are two different delivery models and we presented them as if they were one. The confusion is ours.

The three models, stated separately

	A — BKS-managed single tenant	B — ASTRA-tenanted cloud	C — True on-premise
Infrastructure owned by	BKS	ASTRA	ASTRA
Runs in	BKS AWS account, ASTRA-approved region	ASTRA's own AWS account and region	ASTRA data centre
Application data at rest	BKS-controlled encrypted storage	ASTRA-controlled; ASTRA holds the KMS keys	ASTRA storage
Model inference	Amazon Bedrock in-region, or Anthropic/OpenAI API	Amazon Bedrock in ASTRA's own account and region	Self-hosted open-weight model
Document text leaves ASTRA's cloud boundary?	Yes, to BKS's account	No	No
Leaves the country/region?	No (region-pinned)	No	No
Commercial shape	Subscription	Subscription + ASTRA pays its own cloud bill	Licence + support
Extraction quality	As measured	As measured — same models	Unmeasured. Lower.
Our recommendation	For the POC	For production	Only if policy forbids B

Our recommendation, and the reason

Model B for production; Model A for the POC.

Model B answers your question in the strongest available form: with Amazon Bedrock running in ASTRA's own AWS account in an ASTRA-approved region, **document content never leaves ASTRA's cloud boundary and never leaves the region**. There is no outbound call to Anthropic's or OpenAI's public API, no third-party network hop, and no cross-border transfer. Authentication is AWS IAM against ASTRA's own account — BKS holds no API key that reaches your documents. Bedrock invocations are not retained by AWS and are not used to train the underlying model.

We recommend Model A for the POC only because it is faster to stand up — days, rather than the weeks an account, VPC, IAM and network-approval cycle takes inside ASTRA. If ASTRA's security policy requires the POC itself to run in ASTRA infrastructure, we will run it on Model B and adjust the schedule accordingly.

On Model C, honestly. The application itself runs on-premise without difficulty — it is a single container with file-backed storage, no database server, no message broker, no cache. What does not carry over is the model. A genuinely air-gapped deployment means a self-hosted open-weight model, and **we have not measured extraction accuracy on one**. We are not willing to quote an accuracy figure we have not measured. If Model C is a hard requirement, we should add an open-weight benchmark to the POC and price the outcome afterwards.

What we need from ASTRA to close this

1. Confirmation of the required model (A, B or C).
2. The approved region (we assume UAE / me-central-1 unless told otherwise).
3. Whether ASTRA requires customer-managed KMS keys.
4. Whether ASTRA IT provisions the account and network, or BKS deploys into an account ASTRA provides.

3. AI architecture, data flow, data residency, API costs, model dependency and security controls

3.1 Architecture

Model access sits behind a **single interface with one method**. Every extraction in the system — quotations, technical datasheets, requirements, deviations, meeting minutes, vendor suggestion — calls the same entry point. Behind it are five interchangeable adapters:

Adapter	Authentication	Data path	Status
Amazon Bedrock	AWS IAM, no key	Stays inside the AWS region	Built; live-testing is a POC task
Anthropic	API key	Outbound HTTPS to Anthropic	Built and in production use
OpenAI	API key	Outbound HTTPS to OpenAI	Built; mock-tested, not live-tested
Google Gemini	API key	Outbound HTTPS to Google	Built; mock-tested, not live-tested
Mock	none	No network at all	Used by the entire automated test suite

Two properties of this design bear directly on your query:

- **Prompts are versioned files on disk**, not string literals buried in code. What is asked of the model is a reviewable, diffable artefact. ASTRA can be given read access to the prompt directory as a contractual condition — see §7.
- **Arithmetic stays in Python**. Extractors capture numbers and units *verbatim* from the vendor's document; code decides compliance, normalisation, FX conversion and ranking. **No model is ever asked whether a vendor complies**. This is the most important control in the system: an award recommendation is produced by deterministic, auditable, testable code operating on figures traced to a page — not by a model's judgement.

3.2 Data flow

Diagrammed in Appendix B. Per document:

1. Uploaded over TLS; archive entries validated against path traversal before anything is written to disk.
2. Text extracted **locally** (pypdf / python-docx / openpyxl / poppler). Binaries, images and drawings are not transmitted in the default configuration.
3. Extracted text split on line boundaries into budgeted chunks (12,000 characters technical, 8,000 requirements).
4. Each chunk sent to the configured model with the versioned prompt; returns structured JSON — figures and units, not conclusions.
5. Results merged all-or-nothing and stored locally against the source document, page and vendor.
6. Comparison, compliance checking, normalisation and ranking run entirely in Python over stored facts. **No model call occurs at this stage**.

Volume: approximately **13 MB of text leaves the boundary per 500 MB of source documents** — and under Model B, that boundary is ASTRA's own AWS region.

3.3 Data residency

Data class	Where it lives (Model B)	Crosses a border?
Uploaded documents	ASTRA S3, ASTRA region	No
Extracted facts, comparison results	ASTRA file store, ASTRA region	No
Accounts, sessions, audit trail	ASTRA file store, ASTRA region	No
Bidder registry	ASTRA file store, ASTRA region	No
Text sent for inference	Amazon Bedrock, same ASTRA region	No
Model outputs	Returned to ASTRA region, stored there	No

Training. Under Bedrock's terms, model invocations are not retained and are not used to train models. Under Anthropic's and OpenAI's commercial API terms, prompts and outputs are excluded from training by default. We will attach the current published terms to the contract and re-confirm them at signature, rather than asking you to rely on this paragraph.

3.4 API costs — measured, not estimated

These figures come from replaying a real multi-vendor tender through the production code, not from a model of what it might cost:

Measure	Value	Basis
Model calls per tender	~102	measured — 37 documents, 4 vendors
Wall-clock per tender, today	~34 minutes	measured, fully sequential
Source text yield	18,470 characters per MB	measured, born-digital PDFs
Text amplification (prompt overhead, second passes)	×1.46	measured
Characters per token, this text shape	3.41 in / 2.06 out	measured with the provider's own tokeniser

The assumption most likely to be wrong for ASTRA is the third one. Born-digital PDFs yield 18,470 characters per MB. Scanned drawings yield almost none and cost almost nothing; spreadsheets can yield several times more. **Measuring this against ASTRA's actual document mix is a primary objective of the POC** — it is the number that drives your bill, and we will not contract a per-MB rate before we have measured yours.

One disclosure. Token usage is **not currently metered inside the platform**. The figures above were produced by an offline measurement harness. A usage meter with a per-run cost record is a Phase 1 deliverable (Appendix A, C-6) and must be in place before any consumption-based billing begins. We are telling you this rather than letting you discover that the meter behind the pricing model does not yet exist.

Cost containment. Two levers are available and both are in Phase 1 scope: adopting the provider's Batch API roughly halves inference cost with no change to output, and prompt caching removes repeated cost on the fixed extraction instructions.

3.5 Model dependency

A fair concern, and the design already answers most of it.

- **Switching provider is a configuration change**, not a code change — one environment variable, overridable per ingestion run. The abstraction exists precisely so that no single provider is load-bearing.
- **Under Model B, ASTRA's dependency is on AWS Bedrock**, not on Anthropic or OpenAI as counterparties. Bedrock offers multiple model families, so a model deprecation is a model-id change within an existing contract.

- **Model versions are pinned.** The model id is explicit configuration. ASTRA is never silently migrated to a new model. Any model change is a change-controlled event with a re-run of the accuracy benchmark against the POC corpus.
- **Honest caveat.** The OpenAI and Gemini adapters are built and unit-tested but have not been exercised against live endpoints. If ASTRA wants a contractually guaranteed second provider, live-testing that adapter is a named POC deliverable — not something to assume.

3.6 Security controls

Built and testable today:

Control	Implementation
Authentication	Fail-closed. Every API path requires a session except exactly three (health, login, signup), and that set is pinned by an automated test, so a new endpoint is protected the day it is written.
Passwords	bcrypt with a per-password salt. The digest is structurally incapable of appearing in an API response.
Sessions	Only sha256(token) is stored. Browser-session cookies with no persistence. Restarting the service signs everyone out.
Authorisation	Roles (admin, reviewer) set once at creation; no endpoint can change a role. Per-project grants enforced server-side on every route, not by hiding navigation.
Upload safety	One traversal guard for the whole system, deliberately strict: a . . segment is refused on the segment rather than on where it resolves, and both path separators are normalised first — so an attack valid only on Windows is still refused on Linux. Requirements uploads restricted by extension.
Transport	TLS everywhere, including every model call.
At rest	Managed encryption; customer-managed KMS keys available under Model B.
Audit trail	Every classification, extraction, override, stage transition and re-run recorded with actor and timestamp. RFQ history is append-only — a backward stage transition appends an entry, it never rewrites one.
Secrets	Never baked into an image unless a build flag is passed explicitly; environment files are excluded from the build context.
Concurrency correctness	Every write, <i>and every decision that gates a write</i> , executes inside a single critical section.

Two known exposures, disclosed rather than hidden. Both are development conveniences that must be closed before any ASTRA deployment, and both are named Phase 1 hardening items (Appendix A, C-7):

1. A development authentication bypass exists (off by default) that serves an unauthenticated caller as the first administrator. It must be verified off and the flag removed from production builds.
2. The current sign-in screen carries a demo-account convenience control that embeds a demo password in the browser bundle. It is removed for ASTRA.

Not yet in place, and we would rather say so: SSO / SAML / Entra ID (customisation item C-4), MFA, IP allow-listing, a penetration test report, and SOC 2 attestation for BKS as an organisation. The infrastructure layer's certifications (AWS: SOC 1/2/3, ISO 27001/27017/27018, PCI DSS Level 1) are **AWS's, not ours**, and we will not present them as ours.

4. Customisation cost — deliverables matrix

Accepted

A customisation figure with no deliverables schedule behind it is an invitation to a change-order dispute, and you are right to refuse it. **Appendix A is a fixed-scope matrix:** every deliverable named, quantified with a unit of measure, and either inside the fixed price or explicitly outside it.

The four principles we propose contracting on

1. **Every deliverable carries a countable unit** — "3 RFQ templates", "2 approval workflows", "1 ERP integration, 4 endpoints" — never "workflow configuration".
2. **Anything not named in Appendix A is outside the fixed price**, and both parties can see that at signature rather than at month four.
3. **A change request is priced against a published day rate before work begins**, never invoiced afterwards.
4. **A configuration change is not a change request**. Adding a discipline, a vendor, an RFQ template, a user or a report filter is self-service and free forever. The line between configuration and development is drawn in Appendix A explicitly, because that line is where change-order disputes actually start.

Use-case coverage

Appendix A quantifies Phase 1 as **6 procurement use cases** covered end to end. We propose that the POC (§8) confirms this list against ASTRA's real process before the customisation price is fixed — pricing a workflow build before seeing your actual MR and RFQ flow is how an estimate goes wrong.

5. The 10-named-user limit

Accepted

Ten seats was sized for a pilot — one procurement team proving the tool on real tenders. It is the wrong shape for a platform used across Procurement, Engineering, Estimation and Projects, and holding you to it would produce exactly the outcome neither of us wants: people sharing logins, which destroys the attributed audit trail that is one of the platform's main arguments.

Three structures, for ASTRA to choose

	Option 1 — Named seats	Option 2 — Functional bands	Option 3 — Enterprise
Shape	Per named user	Per department band	Unlimited named users
Best when	Under ~30 users	30–100 users across four functions	Org-wide standard
Included	10 seats, then \$75/user/month (<i>indicative</i>)	Banded at 25 / 50 / 100 seats	All ASTRA staff
Usage allowance	Shared pool — extra seats draw on the same allowance and are never rationed	Shared pool	Shared pool
Our view	Fine for the POC	Recommended for rollout across four functions	If ASTRA standardises platform-wide

Two commitments regardless of which is chosen:

- **An additional seat is never charged twice**. It carries no separate usage allowance; it draws on the account's shared pool, so a user's work is billed once through the meter or not at all.
- **Read-only and occasional users**. Approvers, engineering reviewers and management who read a comparison but never run one should not consume a full seat. We propose a **read-only role at no charge**, capped at a multiple of paid seats, written into the commercial schedule.

Final seat counts should be set from what the POC measures, not guessed now. We would rather size this from your real usage than sell you fifty seats and have thirty-one of them idle.

6. Engineering AI scope

Accepted without reservation

The Engineering AI scope in the proposal is broad, unmeasured and not costed against deliverables. We are **withdrawing it from Phase 1** rather than defending it.

Why

Everything we have quoted an accuracy or cost figure for in this document was measured against a real tender corpus. We have no equivalent measurement for the engineering scope, and a scope with no measurable acceptance criterion cannot be accepted, cannot be tested, and cannot be signed off. Including it would place a large unmeasured item beside a set of measured ones and lend it credibility it has not earned.

Proposed re-scoping

Phase 1 (contracted now): RFQ management and bid evaluation. No engineering AI.

Phase 3 (optional, priced separately once Phase 1 is live): two or three named engineering use cases, each with a written acceptance criterion, each independently cancellable. Candidates, for ASTRA to select and rank:

Ref	Deliverable	Measurable acceptance criterion
E-1	Technical deviation extraction — structured deviation register from vendor technical submittals	≥ X% recall against a hand-built register on N real submittals
E-2	Datasheet parameter extraction — named parameters per equipment class	≥ X% field accuracy on N datasheets, per class
E-3	Specification compliance check — requirement-by-requirement compliance matrix	≥ X% agreement with an engineer's manual matrix
E-4	Drawing / BOM extraction	Not proposed. Requires vision extraction we have not measured.

X and N to be set jointly from the POC baseline. **We will not propose E-4 until we have measured it** — the same discipline that led us to withdraw the broad scope in the first place.

7. Ownership of the solution, source code, data, AI prompts and models — and continuity if BKS support ends

Answerable in full, and we regard the answers below as contractual.

7.1 Ownership

Asset	Owner	Notes
All ASTRA data — documents, extracted facts, comparisons, bidder registry, audit trail, user accounts	ASTRA, unconditionally	Yours during the term and after it. Not conditioned on payment status, notice period or renewal.
ASTRA-specific configuration — RFQ templates, disciplines, approval workflows, report layouts, integration mappings	ASTRA	Delivered as source files, not as opaque database rows.
ASTRA-specific prompts — any prompt tuned to ASTRA's document formats	ASTRA	Versioned files, delivered with the configuration.
Base platform source code	BKS , licensed perpetually to ASTRA for internal use	See 7.3 for the continuity mechanism.
Base platform prompts	BKS, provided to ASTRA under the licence	ASTRA gets read access for audit — a prompt you cannot read is a control you cannot verify.

Asset	Owner	Notes
Underlying AI models	Anthropic / OpenAI / AWS	Neither party owns these; both are third-party services.

ASTRA's data is never used to improve a product sold to anyone else. No ASTRA document, extraction or comparison is used to tune prompts, build benchmarks or train anything used outside ASTRA's own deployment. We will contract on this.

7.2 Exit and migration — the strongest answer we can give you

The architecture makes this straightforward, because there is no proprietary database and no proprietary format anywhere in it:

What you take out	Format	Readable without BKS?
Every uploaded document	Original files, unmodified, in a plain directory tree	Yes — they are your files
Extracted facts and comparison results	JSON, one file per collection	Yes — any text editor
Audit / event log	JSONL, append-only	Yes
Bidder registry	SQLite (open format) + CSV/XLSX export	Yes
RFQ workflow — projects, items, RFQs, shortlists, clarifications	One JSON document	Yes
Accounts and grants	JSON	Yes
Comparisons and compliance matrices	CSV and XLSX export	Yes — Excel
Configuration and prompts	Plain text / YAML files	Yes

There is **no proprietary binary format, no encrypted vendor-locked store, and no data structure that requires BKS software to interpret**. A full export is a copy of a directory. We propose contracting a **quarterly automated export to an ASTRA-owned S3 bucket**, so a current copy is continuously in ASTRA's hands rather than requested at the point of dispute.

7.3 If BKS support is discontinued

Four mechanisms, in ascending order of protection. ASTRA should select the level appropriate to how critical the platform becomes:

1. **Source code escrow** with a recognised agent (NCC or equivalent), released on defined triggers: insolvency, acquisition without novation, or failure to meet the support SLA for a defined period.
2. **Perpetual internal-use licence**, so on termination ASTRA may continue running the version in place indefinitely. Under Model B this is meaningful rather than theoretical — the software already runs in ASTRA's own account.
3. **Documented handover**: deployment runbook, architecture documentation and configuration inventory maintained as a contractual deliverable, rather than produced under exit pressure.
4. **Transition assistance**: a defined number of days' support at a pre-agreed rate during a 90-day transition, whichever party terminates.

One thing worth stating plainly. Under Model B, if BKS disappeared tomorrow, the platform keeps running. It is a container in ASTRA's account, calling Bedrock in ASTRA's account, reading and writing files ASTRA owns. Nothing calls home to BKS and no licence check phones out. What ASTRA loses is future development and support — not access to the system or to the data.

8. The proposed POC — agreed

We agree, and we would have proposed it had you not. This is what the pilot in our proposal was for; your framing is better than ours, because it names the acceptance criteria up front.

Design

Duration	6 weeks — 4 weeks execution, 2 weeks measurement and report
Input	ASTRA's real MR/RFQ packages and vendor quotations — not a demonstration set
Volume	3–5 completed tenders that ASTRA has <i>already evaluated manually</i> , so there is a known-correct answer to measure against
Deployment	Model A (BKS single tenant, ASTRA-approved region) unless ASTRA policy requires Model B
Users	5–8 ASTRA users across Procurement and Engineering
Fee	\$7,500, credited in full against a first-year agreement signed within 60 days of completion
If ASTRA stops at week 6	ASTRA keeps every comparison, every export and the full measurement report. Nothing to unpick.

The seven things you asked us to measure

#	Measure	Method	Proposed pass mark
1	Extraction accuracy	Field by field against ASTRA's own manually built comparative statement	≥ 95% commercial fields; ≥ 90% technical parameters (<i>to confirm</i>)
2	Comparison accuracy	Does the ranked outcome match the answer ASTRA's team reached by hand — and where it differs, which one is right?	100% agreement on the compliant / non-compliant call, or a documented reason for every divergence
3	Time saving	ASTRA records hours spent manually; we record hours on the same tenders through the platform	≥ 60% reduction (<i>to confirm</i>)
4	Security	ASTRA IT/security review against §3.6, plus a documented data-flow walkthrough	ASTRA sign-off, no unresolved high findings
5	Integration effort	Scoping workshop with ASTRA IT on ERP, SSO and mail	A costed integration plan, not an estimate
6	Total cost of ownership	Measured token consumption on ASTRA's own document mix, plus infrastructure, plus support	A 3-year TCO built from measured figures, not from our planning rates
7	Document-mix calibration	Measured characters-per-MB across ASTRA's actual scanned / born-digital / spreadsheet mix	Baseline established — becomes the contractual basis for any usage pricing

Full criteria in Appendix C.

Measure 7 is the one we most want from this POC, because it is the number our entire cost model rests on and we currently hold ASTRA's value at zero. Our planning rate is 18,470 characters per MB, measured on born-digital PDFs. If ASTRA's packages are heavily scanned, that figure collapses and so does the cost; if they are spreadsheet-heavy, it rises. **We are not willing to contract a usage rate against ASTRA's documents before we have measured ASTRA's documents.**

What we need from ASTRA to start

1. Three to five completed, manually evaluated tender packages, with the comparative statement your team produced.
2. The hours those tenders actually took, if recorded — otherwise a considered estimate from the people who did the work.
3. A named ASTRA POC owner and 5–8 users.
4. A security-review slot with ASTRA IT in week 1, not week 6.

5. Agreement on the pass marks above **before** we start, so the result is a fact rather than an argument.

9. Consolidated gap register

Every limitation named anywhere in this document, in one place, so ASTRA can plan against it. This section exists because we would rather lose on an honest scope than win on an implied one.

#	Gap	Impact	Plan
G-1	No mail transport — nothing sends an RFQ, answer or addendum to a bidder	RFQ issue is a manual send today	Customisation item C-3
G-2	No ERP / SAP integration built	Manual PO handoff	Customisation item C-1, scoped in POC
G-3	No SSO / SAML / Entra ID	Separate credentials	Customisation item C-4
G-4	Token usage not metered in-platform	Consumption billing cannot start	Phase 1 deliverable C-6, before any usage billing
G-5	Ingestion runs synchronously — ~34 min per tender, one at a time	Throughput ceiling	Queue + parallel workers, Phase 1. Target ~4–8 min per tender at concurrency 8
G-6	OpenAI and Gemini adapters not live-tested	Second-provider guarantee unproven	Live-test as a POC deliverable if ASTRA wants a contracted alternate
G-7	Bedrock adapter built but not yet wired into a deployed configuration	Model B depends on this	POC deliverable — the first thing we do
G-8	Remote object storage (S3) backend not implemented	Local / EFS storage only today	Phase 1. The storage interface already exists, so this is an addition, not a rebuild
G-9	Development auth bypass and demo-credential control present in the codebase	Must not reach an ASTRA deployment	Phase 1 hardening C-7, verified in the POC security review
G-10	Vision extraction for scanned drawings not enabled	Image-only drawings yield no text today	Deliberate — off by default and costs nothing. Enabling it changes the cost model materially and is a priced decision after the POC measures how much of ASTRA's corpus is scanned
G-11	Cost-estimation module is CLI-only, not in the web application	Not usable by ASTRA today	Out of Phase 1 scope; Phase 3 candidate
G-12	No SOC 2 attestation for BKS as an organisation	ASTRA relies on AWS's certifications plus contractual terms	Disclosed; on the BKS roadmap, not claimed today
G-13	No penetration test report	ASTRA may require one	Can be commissioned before production go-live if ASTRA requires it
G-14	On-premise (Model C) extraction accuracy unmeasured	Model C cannot be quoted with an accuracy figure	Add an open-weight benchmark to the POC only if Model C is a hard requirement

10. What we propose happens next

#	Action	Owner	By
1	ASTRA confirms deployment model (A / B / C) and approved region	ASTRA	Week 1
2	ASTRA confirms whether source-to-pay scope is required, or whether RFQ + evaluation is the target	ASTRA	Week 1

#	Action	Owner	By
3	ASTRA selects seat structure (Option 1 / 2 / 3) for pricing purposes	ASTRA	Week 2
4	ASTRA ranks Phase 3 engineering candidates E-1 to E-3, or defers them	ASTRA	Week 2
5	ASTRA selects continuity level (escrow / perpetual licence / both)	ASTRA	Week 2
6	Joint workshop: agree POC pass marks and finalise Appendix A	Both	Week 2
7	Re-issued proposal — corrected scope, single deployment model, fixed deliverables matrix	BKS	Week 3
8	POC starts	Both	Week 4

Appendix A — Customisation deliverables matrix

To be finalised jointly in the Week 2 workshop. Quantities indicative pending the POC. Once signed, this matrix is the definition of the fixed price.

A.1 Inside the fixed customisation price

Ref	Deliverable	Unit of measure	Qty	Acceptance
Workflows				
C-W1	ASTRA RFQ approval workflow configured to ASTRA's delegation-of-authority matrix	approval workflows	2	A test transaction through each, signed off by the ASTRA process owner
C-W2	ASTRA stage-gate criteria mapped onto the eight built-in stages	stage-gate rules	up to 8	Each gate refuses and permits correctly on a test RFQ
C-W3	Technical bid evaluation workflow with ASTRA's scoring model	evaluation workflows	1	Produces ASTRA's TBE output on a POC tender
Templates				
C-T1	RFQ document templates in ASTRA format	templates	3	Comparable to an ASTRA-issued RFQ
C-T2	Comparative statement matching ASTRA's exact Excel layout	templates	1	Opens in ASTRA's own CS workbook without rework
C-T3	Clarification / query register template	templates	1	ASTRA sign-off
C-T4	Technical bid evaluation report template	templates	1	ASTRA sign-off
Integrations				
C-1	ERP integration — vendor master + PR/PO	endpoints	4	Round-trip test against ASTRA's non-production ERP
C-3	Outbound mail transport — RFQ issue, clarification answers, addenda	integrations	1	RFQ issued to a test mailbox with a delivery record
C-4	SSO — SAML 2.0 / Entra ID	integrations	1	ASTRA credential signs in; local passwords disabled
C-5	Vendor master import from ASTRA's AVL	importers	1	Full ASTRA AVL loaded, row count reconciled
Reports				
C-R1	Standard reports — spend by discipline, cycle time by stage, vendor participation, clarification ageing	reports	4	Each renders on POC data and exports to XLSX
C-R2	Management dashboard	dashboards	1	ASTRA sign-off
Platform				
C-6	Token/usage metering with per-run cost record and threshold alerts	1	1	Meter reconciles to the provider invoice within 5%
C-7	Security hardening — dev bypass removed, demo control removed, production build verified	1	1	Verified in the ASTRA security review
C-8	Queue + parallel workers (throughput)	1	1	A tender completes in ≤ 8 minutes at concurrency 8
Enablement				
C-E1	Administrator training	sessions	2	Attendance + an admin task completed unaided
C-E2	End-user training	sessions	4	Attendance + a tender run unaided
C-E3	Documentation — admin guide, user guide, runbook	documents	3	ASTRA sign-off

Use cases covered end to end by the above: 6 — (1) raise an RFQ from an MR; (2) shortlist from the AVL with approval; (3) issue an enquiry package; (4) run a clarification round; (5) receive and compare bids; (6) produce the TBE and comparative statement for award.

A.2 Free configuration — never a change request

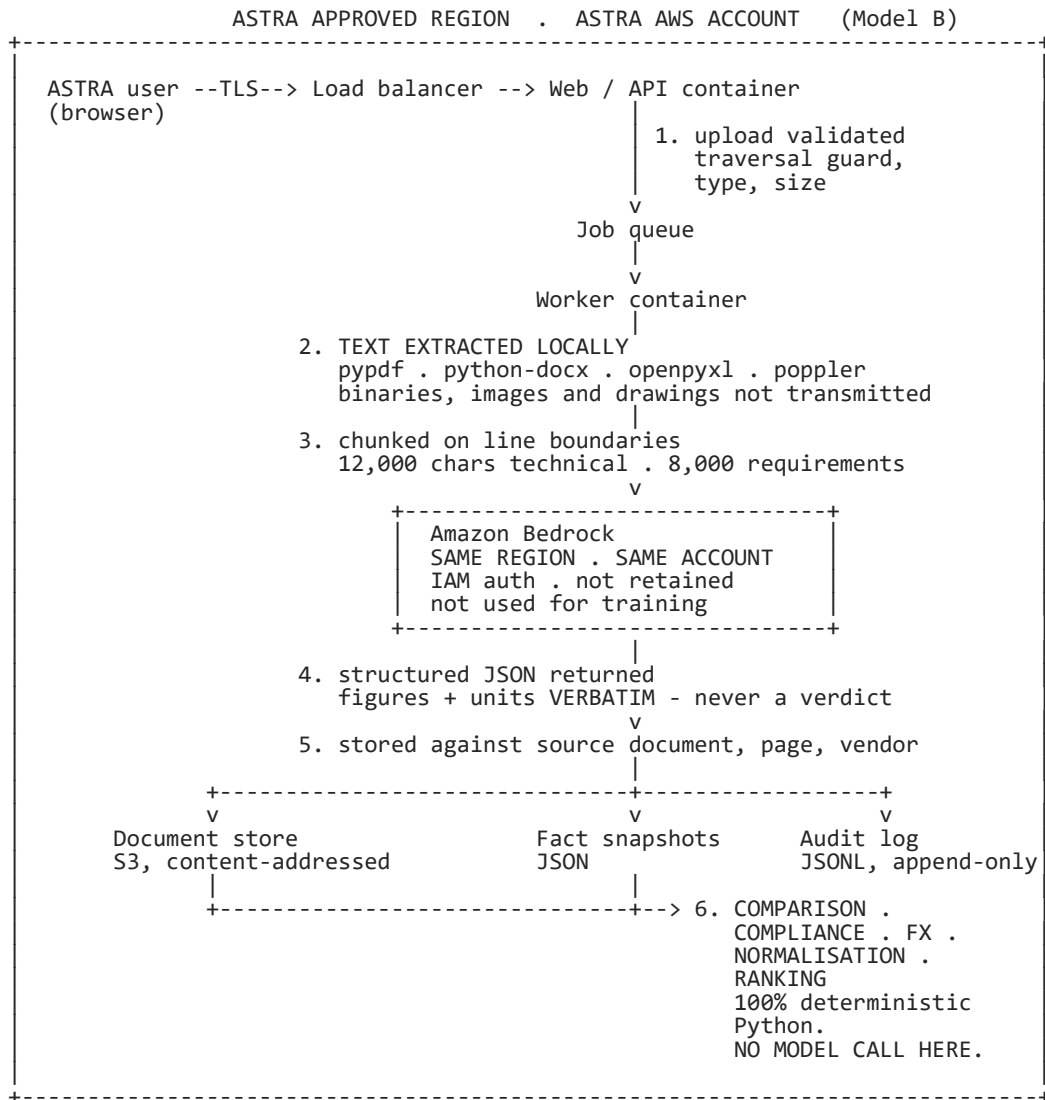
Adding or editing: disciplines and product groups · vendors and vendor contacts · users, roles and project grants · RFQ metadata fields · report filters and saved views · approval thresholds within an existing workflow · currencies and FX rates · document classes and file-naming rules · email recipient lists.

A.3 Explicitly outside the fixed price

Additional ERP endpoints beyond the 4 in C-1 · additional integrations beyond C-3 to C-5 · a second ERP · custom mobile applications · contract lifecycle management · invoice matching or payment · e-auction · supplier performance scorecarding · catalogue / punch-out · engineering AI use cases (§6, Phase 3) · migration of historical tenders predating go-live · custom reports beyond the 4 in C-R1.

Each priced against a published day rate, quoted and approved **before** work begins.

Appendix B — Data flow



Nothing crosses this boundary. No call to BKS. No call to a public model API.

Under Model A, the boundary is a BKS-controlled account in an ASTRA-approved region, and steps 3–4 may run against Amazon Bedrock (in-region) or against the Anthropic API (outbound; no training, no retention beyond abuse screening) — ASTRA's choice, recorded in the contract.

Appendix C — POC acceptance criteria

Pass marks proposed. To be agreed in the Week 2 workshop before the POC starts, so that the result is a measurement rather than a negotiation.

#	Criterion	Method	Proposed pass	Evidence produced
1a	Commercial field extraction accuracy	Field by field vs ASTRA's manual comparative statement, all POC tenders	≥ 95%	Per-field diff report, every divergence listed
1b	Technical parameter extraction accuracy	Vs ASTRA's manual technical comparison	≥ 90%	Per-parameter diff report
1c	Document classification accuracy	Vs manual classification of every file in the POC corpus	≥ 95%	Confusion matrix
1d	Revision lineage correctness	Every superseded document correctly retired	100%	Lineage report
2a	Ranked outcome agreement	Platform ranking vs ASTRA's manual award recommendation	Match, or a documented reason per divergence	Side-by-side ranking
2b	Compliance determination	Compliant / non-compliant call per vendor per requirement	100% agreement, or a documented reason	Compliance matrix vs ASTRA's
2c	Traceability	Every figure traceable to source document and page	100%	Spot-check of 50 random figures
3	Time saving	ASTRA-recorded manual hours vs platform hours, same tenders	≥ 60% reduction	Time log, both methods
4a	Security review	ASTRA IT review against §3.6 + data-flow walkthrough	Sign-off, no unresolved high findings	Review record
4b	Data residency verification	ASTRA verifies no egress outside the approved region	Verified by ASTRA	Network / flow evidence
5	Integration effort	Workshop with ASTRA IT on ERP, SSO, mail	A costed plan, not an estimate	Integration plan with day counts
6	Total cost of ownership	Measured tokens on ASTRA's mix + infrastructure + support	3-year TCO from measured figures	TCO model, every assumption sourced
7	Document-mix calibration	Characters-per-MB across ASTRA's actual scanned / born-digital / spreadsheet mix	Baseline established	Calibration report — the contractual basis for usage pricing
8	Throughput	Wall-clock per tender under production configuration	≤ 8 min at concurrency 8	Timing log

If criteria 1, 2 and 3 are not met, ASTRA does not proceed and owes nothing further. The \$7,500 covers the work either way, and ASTRA keeps every output and the full measurement report. We would rather find that out in six weeks on five tenders than in year one on five hundred.

Prepared by BKS Solutions in response to ASTRA's review of the Procurement Comparison Portal proposal, 18 August 2026. Technical statements in §1, §3 and §7 describe the platform as built and are verifiable during the POC. Commercial figures marked indicative are subject to the commercial schedule.